Sunwoo Hong

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Research Interests

I work on **closing the gap between what people do easily and what models still cannot**: people reason instead of matching patterns, learn from very little, and act in a physical world they understand. I currently focus on how robots **judge and control their own actions**: learning that judgment from their own practice rather than imagined futures, and layering robot control like a nervous system. My approach is to find the assumption that holds a model class back and replace it.

Keywords: world models, embodied AI, robot learning, VLA models, reasoning, diffusion language models, data-efficient learning

Education

Korea Advanced Institute of Science and Technology (KAIST) — Seoul, South Korea
M.S. in Artificial Intelligence, Graduate School of AI — *Feb 2027 (expected)*
Advisor: Prof. Se-Young Yun

Seoul National University (SNU) — Seoul, South Korea
B.S. in Electrical and Computer Engineering — *Feb 2024*

Sejong Academy of Science and Arts — Sejong, South Korea
Inaugural Student Council President; Government-designated academy for gifted students — *Feb 2018*

Publications

∗: equal contribution (random order), †: corresponding author, C: conference, W: workshop, U: under review, P: in preparation, D: domestic

[C2] S. H. Kim*, **Sunwoo Hong***, Y. Choi, C.-H. Chao, S.-Y. Yun†, and R. G. Krishnan†. Pivot-SD: Efficient Self-Distillation for Masked Diffusion Language Models. **Empirical Methods in Natural Language Processing (EMNLP), Main Conference**, 2026. **Oral presentation.** [project]

[C1] S. H. Kim*, **Sunwoo Hong***, H. Jung, Y. Park, and S.-Y. Yun. KLASS: KL-Guided Fast Inference in Masked Diffusion Models. **Neural Information Processing Systems (NeurIPS)**, 2025. **Spotlight (top 3.2% of submissions).** [pdf] [code]

[U1] **Sunwoo Hong***, S. H. Kim*, C.-H. Chao, Y. Choi, S.-Y. Yun†, and R. G. Krishnan†. Logical Equivariance in Diffusion Language Models. Under review at **ICLR 2027**. Earlier version presented at **ICML Workshop** on Foundations of Deep Generative Models, 2026. [pdf]

[P1] **Sunwoo Hong**, A. Pesaranghader†, and S.-Y. Yun†. World Models Need Judgment. In preparation for submission to CVPR 2027.

[P2] **Sunwoo Hong**, A. Pesaranghader†, and S.-Y. Yun†. Neuraxis: From Dual-System to Nervous-System Control in Vision-Language-Action Robots. In preparation for submission to ICML 2027.

[W1] C. Park, **Sunwoo Hong**, K. Kim, Y. Park, A. W. Olson, and A. Pesaranghader†. Route-Guided CBF-QP Repair for Safe Vision-Language-Action Models. **RSS Workshop** on Trustworthy Embodied Foundation Models, 2026. [pdf] [project] [code]

[D1] **Sunwoo Hong**, Y. Kim, S. Jung, J. Oh, and S.-Y. Yun. Memory Reduction via Ternary Quantization in Reinforcement Learning. Korea Institute of Military Science and Technology (KIMST) Conference, 2025.

Research Experience

Graduate Research Assistant, OSI Lab, KAIST AI — *Feb 2025 – Present*
Advisor: Prof. Se-Young Yun [page]

- Showed that world models lack the judgment decisions need: a 2B video world model's futures pick the best push at chance, yet a grasp is judged from its recording at AUC 0.85 with 16 labels (0.70 beforehand with 158). Read-to-Verify learns it from the robot's practice, nearly matching 158 labeled episodes with 16 [P1].
- Built a nervous-system control stack for VLAs, where a planner writes checkable contracts and fast layers check them every action chunk. Proved that standard data lets scene and posture override the instruction [P2].
- Found that the KL divergence between consecutive denoising steps tells a masked diffusion model which tokens are already settled. Unmasking those in parallel, with no training, gives up to 2.78× faster generation while improving on greedy decoding [C1].

Visiting Researcher, University of Toronto & Vector Institute *Jan 2026 – Jul 2026*
Advisor: Prof. Rahul G. Krishnan [page]

- Showed that diffusion language models solve about 99% of Sudoku states in at least one of eight rotated or reflected views but only 13–24% in all eight, and proved that hard-wiring this symmetry into attention leaves it almost no capacity, so consistency must be learned [U1].
- Found that a few early commitments decide most of a diffusion language model's output. Training on those alone (10 tokens per trajectory, under 4%) beats full-sequence fine-tuning and RL runs with ten times the data, at one-sixth to one-eighth of the wall-clock time [C2].

Research Collaborator, LG Electronics Toronto AI Lab *Jan 2026 – Jul 2026*
Mentor: Dr. Ali Pesaranghader [page]

- Safety filters keep a robot policy from colliding but push it off its task path. Co-developed a module that routes it around the obstacle and back, improving collision avoidance and task success over AEGIS on SafeLIBERO [W1].

Graduate Research Assistant, MLML Lab, KAIST School of Computing *Feb 2024 – Jan 2025*
Advisor: Prof. Sungjin Ahn [page]

- Designed a compact 1D tokenizer that compresses video into a short sequence of global tokens instead of a spatial grid, to model scene-level dynamics in video world models.
- Studied why block-slot object-centric models fail to separate factors such as shape and color, and built out-of-distribution tests that measure compositional generalization.

Undergraduate Researcher, AISys Lab, SNU *Jan 2023 – May 2023*
Advisor: Prof. Jinho Lee (B.S. thesis) [page]

- Built pipeline-parallel training schedules for knowledge distillation that remove idle time between teacher and student passes: up to 1.45× faster training or 1.55× better memory efficiency than a GPipe baseline.

Research Intern, Human-Centered Computer Systems Lab, SNU *Jun 2022 – Sep 2022*
Advisor: Prof. Youngki Lee [page]

- Built a server-driven adaptive video streaming pipeline for deep learning inference, reducing bandwidth by up to 70% while maintaining detection accuracy.

Undergraduate Research Program (URP), SNU *May 2021 – Dec 2021*
Advisor: Prof. Kyomin Jung [page]

- Developed a GAN-based model that transfers a sign-language interpreter's motion, extracted as skeletons, onto the speaker in a video, so that the speaker appears to sign.

Awards and Honors

- Oral Presentation (2.9% of submissions), EMNLP 2026 *Oct 2026*
- Excellence Selection, GARL 2026 Challenge (National AI Research Lab, Korea) [U1] *Jul 2026*
- Spotlight (top 3.2% of submissions), NeurIPS 2025 *Dec 2025*
- Full Scholarship, Choi Hyuk-Young Scholarship Foundation *Mar 2019 – Present*
- Minister of Education Award (1st place), national STEAM R&E research competition *Nov 2015*

Teaching and Academic Service

- Teaching Assistant, Geometric Deep Learning (AI810), KAIST *Spring 2025*
- Teaching Assistant, Reinforcement Learning (CS672), KAIST *Fall 2024*
- Teaching Assistant, Freshman Seminar: Writing (E51.101), Seoul National University *Fall 2021*
- Reviewer, ICLR 2027 and ACL Rolling Review (August 2026 cycle) *2026*

Military Service

- Republic of Korea Air Force, Sergeant (honorable discharge); information systems specialist *Aug 2019 – May 2021*